Lab 4 — Inverses, Rank, and Determinants

Friday, November 6 (~25 min)

Last updated: November 4, 2026.

In class Friday, November 6 (same day as Lecture 25 / Quiz 9). Quiz 9 is Lecture 22 only — **not** this lab. Today's lab practices Lecture 24 (inverses / rank / null space) and Lecture 25 (determinants) in software.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Learning objectives

1. Compute a matrix inverse in software and check $A^{-1}A = I$.
2. Recognize failure of `inv` / a singular matrix, and connect that to a nontrivial null space.
3. Read rank and nullity ($\dim N(A) = n - \text{rank}(A)$) from software or RREF ideas.
4. Compute determinants and interpret $\det A \neq 0$ vs $\det A = 0$.

Math connections

Lecture 24 and Lecture 25. Good Midterm 2 practice for inverses, rank, and determinants.

Key syntax (this lab)

Task	Octave	Python
Inverse	`inv(A)`	`np.linalg.inv(A)`
Determinant	`det(A)`	`np.linalg.det(A)`
Rank	`rank(A)`	`np.linalg.matrix_rank(A)`
Null space (optional)	`null(A)`	`scipy.linalg.null_space(A)`
Check	`Ainv*A`	`Ainv @ A`

(If `null` / SciPy is unavailable: solve $Ax = 0$ by hand or with RREF for the singular example.)

Ops strip — an invertible 2×2

Use the Lecture 24 example

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}.$$

C4.0. Enter A. Compute A^{-1} in software. Print $A^{-1}A$ (or AA^{-1}). What should you see?

Expected: $A^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$; product is (numerically) I.

C4.1. Check against the 2×2 formula

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Does software match? What is $\det A$?

Expected: $\det A = 1$; formula matches C4.0.

Application — singular matrix, rank–nullity, determinants

C4.2. Let

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

(Lecture 24's non-invertible example). Try `inv(B)`. What happens (error, warning, or huge numbers)?

Expected: failure / singular warning — B has no inverse.

C4.3. For the same B: find $\text{rank}(B)$ and a nonzero vector in $N(B)$ (hand or software). How many free variables? Check rank–nullity: $\dim N(B) = 2 - \text{rank}(B)$.

Expected: $\text{rank}(B) = 1$, nullity 1, e.g. $(2, -1)^T$ (or $(-2, 1)^T$) in $N(B)$.

C4.4. Determinants and meaning.

(a) Compute $\det A$ and $\det B$ (hand or software).

(b) One sentence each: what does $\det A \neq 0$ tell you? What does $\det B = 0$ tell you?

(c) (Optional stretch.) $\det \begin{bmatrix} 2 & 0 & 0 \\ 1 & -3 & 0 \\ 4 & 5 & 1 \end{bmatrix}$ — use the triangular shortcut.

Expected: (a) $\det A = 1$, $\det B = 0$. (b) A invertible / full area scale; B singular, image of the unit square has area 0, no B^{-1}. (c) product of diagonal $= 2 \cdot (-3) \cdot 1 = -6$.

One-pager

Preview the Lab 4 ticket; write on the hard copy in class.